calm_rank_tilt_legacy

equity strategy | family: equity-volatility | status: reference | derived from `calm_rank_tilt`

What this strategy does

Balanced Leverage, with bets shrunk sharply for the jumpiest stocks (ranked against last quarter) -- ORIGINAL leverage cap (new positions only, open book ignored)

Why it should work. PEAD alpha per dollar is highest in calm stocks; rank-weight toward them without depending on vol levels.

Design notes (from the strategy's source file).

Calm Rank Tilt -- Balanced Leverage, tilted hard toward the calmest stocks by volatility RANK (research).

Calm Weighted sized each event by 1/idiosyncratic-vol (pre-fix Sharpe 1.30 -> 1.47; corrected run 1.09 -> 1.24). The attribution work showed that PEAD alpha per dollar is concentrated in low-volatility names and almost absent in the wildest tercile, and that the 1/vol weight, being linear in the vol LEVEL, only mildly demotes the high-vol tail. Weighting by (1.02 - vol percentile)^2 -- where the percentile is measured against the previous quarter's cross-section within the stock's size quintile -- demotes it hard while staying smooth:

Corrected run: Sharpe 1.09 (Balanced Leverage) -> 1.24 (Calm Weighted) -> 1.47; max drawdown -18.6% -> -17.5% -> -13.6%; every era improves (pre-fix: 1.30 -> 1.47 -> 1.81, -16.6% -> -11.6%) split-sample check: exponent picked on 1996-2004 data alone (best k ~ 3) lifts 2005-2013 Sharpe 0.93 -> 1.10 (pre-fix, not re-run), the same as the full-sample best, so the gain is not an artifact of one period.

Lower volatility buys room: `calm_rank_levered` spends it on 1.7x the base unit and a 4x gross cap for ~9% a year at ~5% vol (a market-neutral book clearing the S&P 500's return in this sample). More sizing than that stops paying: the 5%-of-ADV liquidity cap starts to bind and 2008-13 Sharpe decays.

STATUS research: the exponent and sizing were chosen after seeing Calm Weighted's full-sample results (the split-sample check above is the only out-of-sample evidence), and the levered variant leans on small-cap capacity.

parameter	value
base_unit_fraction	0.003
max_gross_leverage	2.5
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	priority
exposure_cap	entry_cohort
beta_overlay	0.0

Performance

metric	calm rank tilt legacy	market (CRSP VW)
Annualized return	7.12%	9.94%
Annualized volatility	3.83%	19.35%
Sharpe (NAV-pct)	1.81	0.59

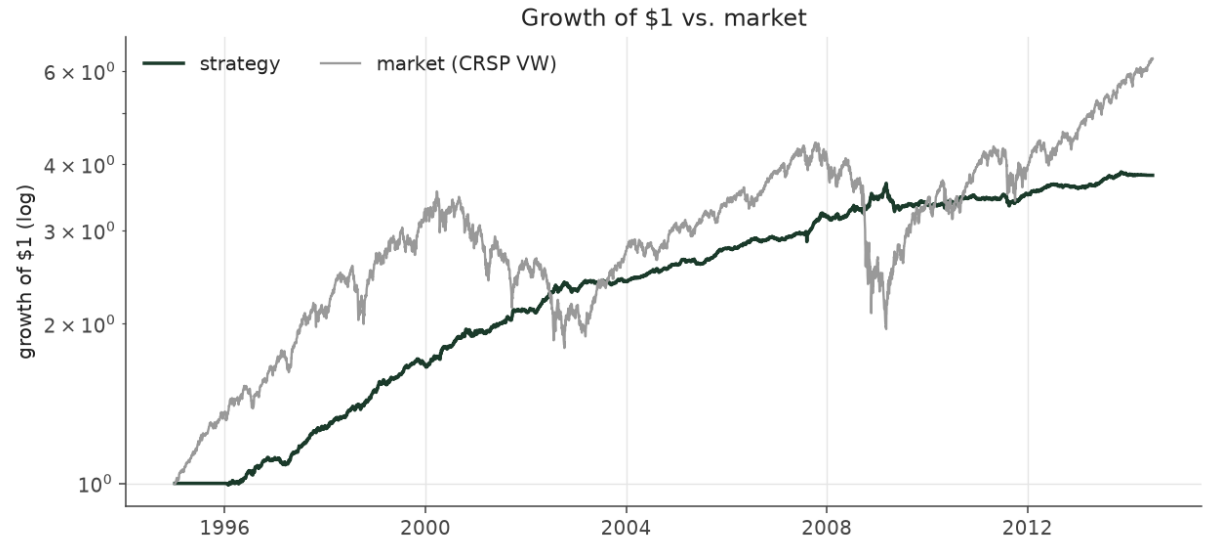
Sortino	2.77	
Max drawdown	-11.6%	-55.5%
Calmar	0.61	
Final NAV ($10M start)	$38,141,081	
Worst day	-2.55%	
% positive days	52.7%	
Sharpe 1996-2001	2.94	
Sharpe 2002-2007	1.89	
Sharpe 2008-2013	0.78	

Correlation of daily returns with the market: **-0.01**.

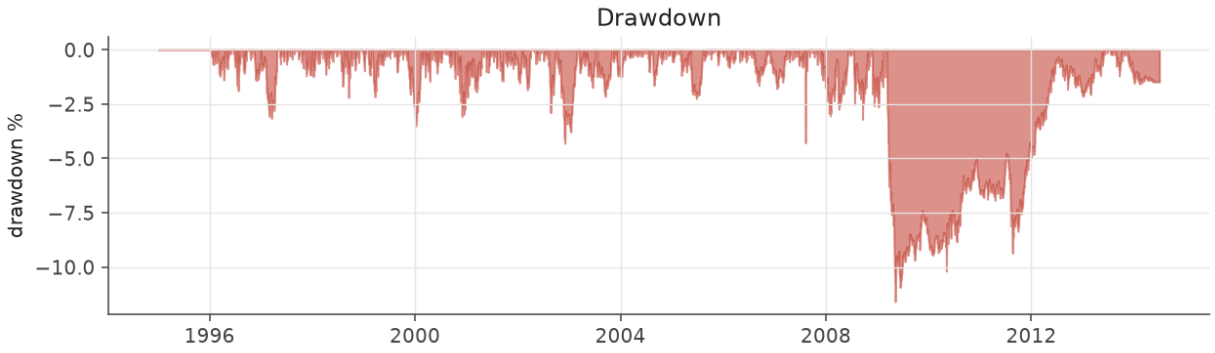
• best Sharpe among equity: `calm_rank_tilt` (1.47); this one: 1.81

• best return among equity: `market_boost_max` (0.116); this one: 0.0712

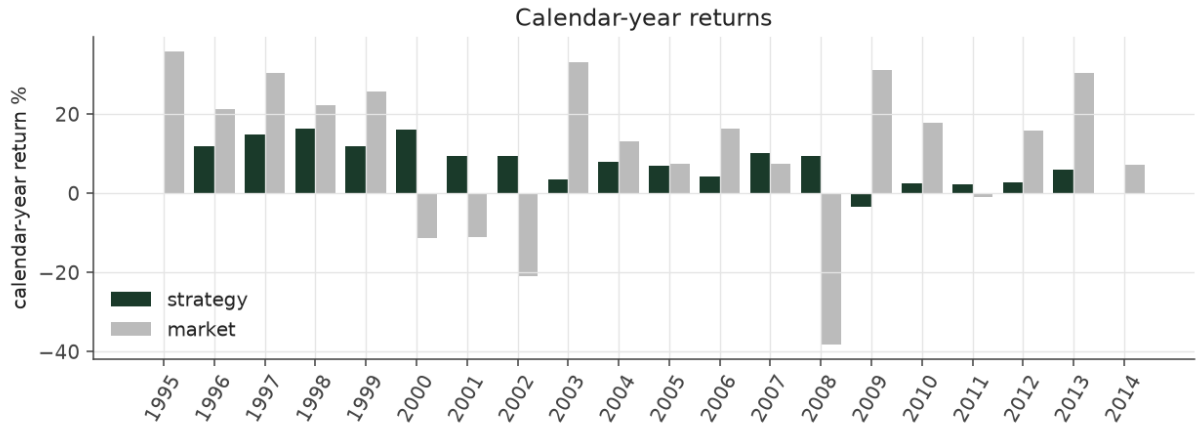
• best Calmar among equity: `calm_rank_tilt` (0.442); this one: 0.614



NAV (log scale)



Drawdown from running peak

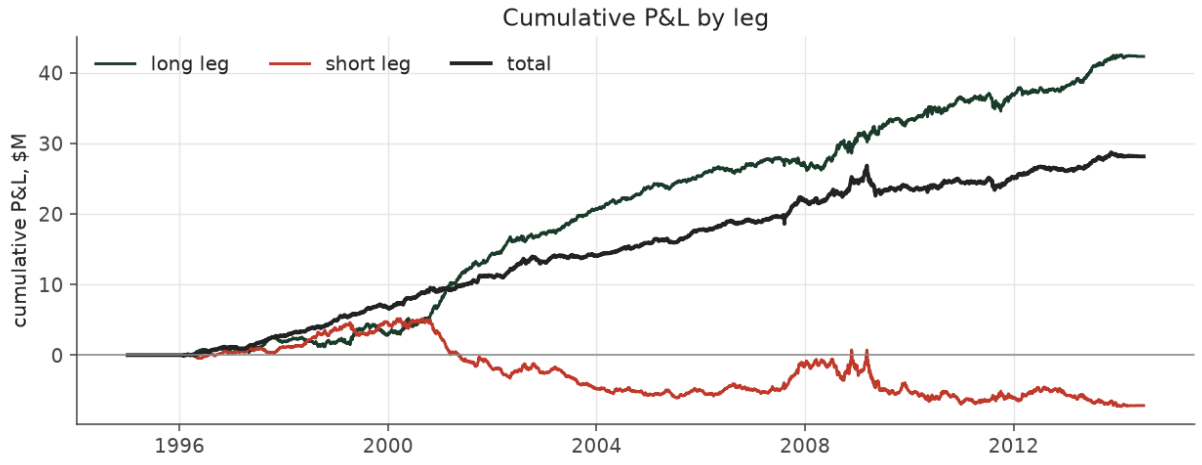


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	42.31	-7.14
P&L after costs ($M)	38.92	-10.78
of which common (average-event) component ($M)	27.21	-27.43

of which excess / PEAD component ($M)	14.63	20.63
excess minus costs ($M)	11.26	17.01
gross return per $ traded (bp)	244.60	-40.33
positions in the money (hit rate)	0.56	0.49
stand-alone Sharpe (leg only)	1.48	-0.16

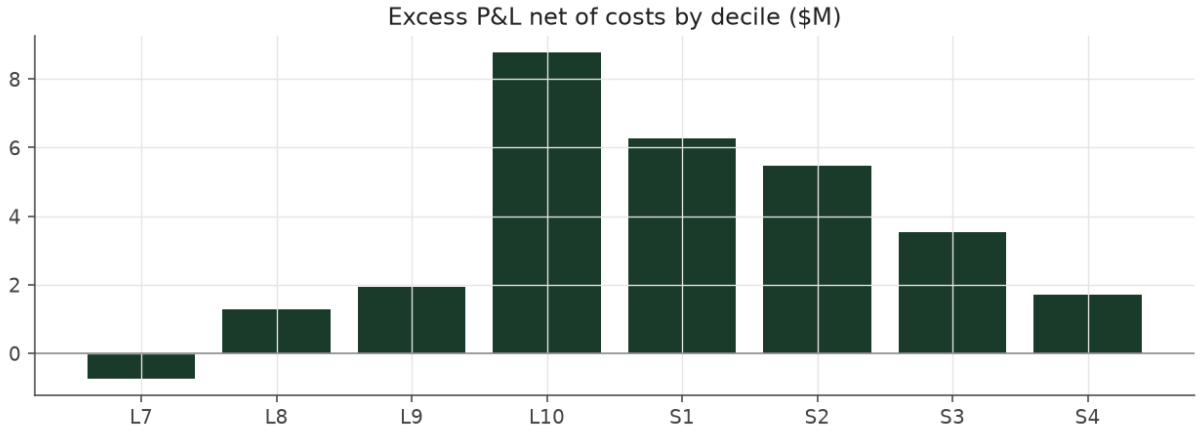
Daily P&L correlation between the two legs: **-0.79**.

• **The short leg loses money in absolute terms** (-7.1M before costs) while the long leg earns +42.3M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 27.2M / 27.4M. Stripping that common component, the surprise signal itself earned +14.6M on the long side and +20.6M on the short side.

• The two legs' daily P&L are -0.79 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.48 long / -0.16 short vs 1.81 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	8,029	158.66	2.31	-0.44	0.31	-0.75	-2.6%	-47.07
long	8	16,588	415.11	8.24	2.10	0.82	1.29	4.5%	30.98
long	9	17,747	535.73	9.62	2.97	1.03	1.93	6.8%	36.09
long	10	17,174	620.10	18.75	10.00	1.22	8.78	31.0%	141.52
short	1	17,169	611.53	-2.46	7.59	1.31	6.27	22.2%	102.54
short	2	17,281	543.16	-3.69	6.58	1.11	5.48	19.4%	100.81
short	3	15,970	433.58	-3.41	4.41	0.87	3.54	12.5%	81.67
short	4	9,171	182.37	-1.23	2.05	0.35	1.70	6.0%	93.45



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	14,213	258.09	5.07	6.75	1.01	5.74	20.3%	222.50
2	22,885	608.55	6.10	7.95	1.76	6.19	21.9%	101.79

3	25,501	801.21	7.23	8.64	1.75	6.90	24.4%	86.10
4	27,583	905.05	6.03	7.29	1.48	5.81	20.5%	64.16
5	28,947	927.33	3.71	4.63	1.03	3.60	12.7%	38.79

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	13,650	208.10	4.29	5.69	1.04	4.65	16.4%	223.25
2	23,084	716.85	6.64	9.78	2.15	7.63	27.0%	106.46
3	25,851	912.90	8.09	9.95	1.83	8.12	28.7%	88.98
4	27,665	861.29	5.41	5.80	1.21	4.60	16.3%	53.37
5	28,874	800.98	3.71	4.03	0.80	3.23	11.4%	40.28

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	45,855	1,393.64	2.33	9.90	2.49	7.41	26.2%	53.15
41-60d	25,652	807.71	6.04	5.77	1.65	4.12	14.6%	51.01
61-90d	35,324	966.11	8.08	9.17	2.05	7.12	25.2%	73.66
>90d	12,298	332.76	11.69	10.43	0.83	9.60	33.9%	288.35

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	33,895	660.14	11.04	12.39	1.31	11.08	39.2%	167.87
2002-07	39,956	1,201.49	10.83	12.96	2.37	10.60	37.5%	88.18
2008-13	45,278	1,638.60	6.27	9.91	3.35	6.56	23.2%	40.05

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	22,094	348.54	3.31	3.99	0.76	3.23	11.4%	92.82
Chems	2,590	232.45	1.79	2.07	0.40	1.67	5.9%	71.87
Durbl	2,843	247.37	1.98	2.44	0.44	1.99	7.0%	80.56
Enrgy	6,194	273.83	3.43	4.03	0.51	3.51	12.4%	128.29
Hlth	11,696	332.40	2.17	2.76	0.69	2.07	7.3%	62.28
Manuf	12,034	318.17	2.01	2.73	0.66	2.07	7.3%	64.96
Money	17,583	315.04	2.93	3.77	0.71	3.06	10.8%	97.27
NoDur	5,081	282.00	1.73	2.20	0.53	1.67	5.9%	59.07
Other	18,377	338.29	3.08	3.97	0.76	3.20	11.3%	94.65
Shops	11,984	326.93	3.01	3.61	0.68	2.93	10.4%	89.71

Telcm	3,981	259.18	1.94	2.36	0.46	1.90	6.7%	73.13
Utils	4,672	226.03	0.76	1.34	0.41	0.93	3.3%	41.15

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	39,523	293.11	3.29	4.03	0.55	3.48	12.3%	118.82
low vol	40,091	1,985.36	14.71	19.34	4.09	15.25	53.9%	76.81
mid vol	37,903	1,165.50	9.67	11.48	2.26	9.22	32.6%	79.12

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	7.12%	-0.00%
0.5%	1.8	6.85%	0.26%
1.0%	3.5	6.58%	0.53%
2.0%	7.1	5.99%	1.12%
5.0%	17.7	3.74%	3.37%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.